Contents

Project: Azure Capacity Reservation Management Engine (ACRME)
Classification: Principal Cloud Architect — Architecture Governance
Version: 1.1
Date: August 2026
Status: Accepted
Part of: ACRME Architecture Decision Records — this is one of four standalone ADRs split from the consolidated ADR set.

> **About ADRs.** An Architecture Decision Record captures a single significant architectural decision, the context that forced it, the options considered, the choice made, and its consequences. ADRs are immutable once accepted — a superseding decision is recorded as a new ADR rather than editing the original. Evidence tags: `[Documented]`, `[Decided]`, `[Derived]`, `[Assumed]` (see Appendix).

ADR-004 — Forecast and Increase of Capacity and Quota

Status: Accepted
Date: August 2026
Deciders: Principal Cloud Architect, Platform Engineering, FinOps, Capacity Planning
Related decisions: D10; FR-7, FR-4.4; G-24
Source: `azure_cr_management_engine_design.md` FR-7; `acrme_production_readiness_review_and_architectu`
§Forecast/§30

Context

Reserved capacity must grow ahead of demand — but Azure quota increases take time to approve, and over-reservation wastes money. The engine needs to forecast demand, recommend right-sized capacity, and drive quota/CR increases with enough lead time, **without** ever autonomously performing material or destructive changes in Phase 1. This growth path must be architecturally distinct from crisis operations (ADR-003).

Decision

Adopt **forecast-driven, approval-gated capacity growth** operating exclusively in `STEADY_STATE`:

1. **Forecasting** analyses historical CR allocation and projects demand over a configurable window (default 30/60/90 days), exposing raw time series and derived recommendations via API. `[Documented - FR-7.1/7.6]`

2. **Capacity sizing formula:** `[Documented - FR-7.3]`

   ```
   Forecast_Quantity = ceil(Forecast_Peak × (1 + Growth_Buffer) + DR_Buffer)
   ```

   Increase when forecast demand exceeds current reserved quantity within the window; right-size down when demand is consistently below current, honouring a configurable buffer.

3. **Lead-time alerting:** when forecast demand approaches a quota limit (default **80%**), emit `ForecastApproachingQuotaLimit` with a **14-day lead time** — enough for quota-increase processing. `[Documented - FR-7.4]`

4. **Auto-increase is approval-gated in Phase 1.** The trigger uses utilisation thresholds with **debounce/cooldown** to avoid thrashing; a `CapacityIncreaseRequest` entity carries the full lifecycle (create → approve → execute → retry → cancel), and Phase 1 requires **operator approval** before execution. `[Decided - D10; Derived - §30, G-24]`

5. **Quota increases** are initiated via the Azure Support REST API (`Microsoft.Capacity/.../serviceLimits`) — at group level where Quota Groups apply (ADR-002) — subject to the same operator-approval gate. `[Documented - FR-4.4]`

6. **Mode isolation.** Auto-increase runs **only** in `STEADY_STATE` and is **suppressed during** `DR_EVENT_ACTIVE`, keeping organic growth strictly separate from crisis transfer (ADR-003). `[Decided - D10]`

7. **Non-destructive guarantee.** Increases only ever raise CR quantity or request more quota; guarded reduction (right-sizing) never drops a CR below its allocated count (platform floor, FR-1.6) and is itself approval-gated.

Steady-State Capacity Lifecycle (normative 10-step policy) The steady-state increase runs strictly separate from DR crisis operations and follows a fixed sequence, with **no assumed quota-propagation SLA**: `[Documented - §30]`

1. Detect threshold crossing.
2. Re-read current CR, quota, sharing, and assignment state.
3. Create `CapacityIncreaseRequest`.
4. Calculate target quantity (`Forecast_Quantity` formula above).
5. **Require operator approval (Phase 1).**
6. Submit the quota action only if validated as required.
7. Wait for confirmed quota state **without assuming a propagation SLA**.
8. Update CR quantity.
9. Confirm the actual quantity.
10. Refresh the snapshot and close the request.

Auto-Decrease Exclusion Auto-decrease is **excluded from Phase 1**: it can remove future capacity and interact with running VMs. Right-sizing down remains operator-driven and guarded by the platform floor (never below allocated count). `[Documented - §30]`

Forecast Advisory Posture Forecast recommendations stay **advisory until model accuracy and false-positive rates are measured**; the horizon is 30/60/90 days and `Growth_Buffer`/`DR_Buffer` are policy percentages, not fixed constants. `[Documented - §28]`

Consequences

Positive: - Capacity grows ahead of demand with sufficient lead time for quota approvals — reducing capacity-exhaustion incidents. - The debounce/cooldown and approval gate prevent runaway or thrashing increases. - `CapacityIncreaseRequest` gives a fully auditable, retryable, cancellable growth workflow. - Right-sizing recovers cost from over-reserved CRs without risking allocated VMs.

Negative / trade-offs: - Approval-gated in Phase 1 means growth is not instantaneous — acceptable because Emergency Transfer (ADR-003) covers crisis speed. - Forecast accuracy depends on history; workload-tagged per-workload forecasts (FR-7.5) are only partially covered (gap G-16). - Threshold, buffer, and cooldown values are empirical and require tuning during the pilot; SLA/propagation times are measured, not invented. - `CapacityIncreaseRequest` lifecycle (entity, approval, retry, cancellation) is still a backlog item (G-24) requiring an end-to-end approved-increase test.

Alternatives Considered

Alternative	Why Rejected
Fully autonomous auto-increase	Removes human control over material cost/quota changes; unacceptable in Phase 1. `[§30]`
Auto-increase escalating into emergency tiers	Conflates growth with crisis response; could run emergency ops without a DR declaration. `[D10]`
Fixed-timer cooldown	Less adaptive than recovering the budget incrementally using observed success rates. `[PRR §-]`
No lead-time alerting (react on exhaustion)	Quota approvals take too long; reacting at exhaustion guarantees deployment failures. `[FR-7.4]`
Sizing without a DR buffer	Under-reserves relative to DR obligations; the formula includes an explicit `DR_Buffer` term. `[FR-7.3]`

Appendix — Decision Log Cross-Reference

ADR	Primary Decisions	Hard Constraints	Key Gaps/Blockers
ADR-004 Forecast & Increase	D10	— (uses HC-3 at execution)	G-16 (workload tags), G-24 (increase entity)

Appendix — Status Legend

Status	Meaning
Proposed	Under discussion; not yet ratified
Accepted	Ratified and in force
Deprecated	No longer recommended but not yet replaced
Superseded	Replaced by a later ADR (referenced explicitly)

Appendix — Evidence Tag Taxonomy

Tag	Meaning
`[Documented]`	Traceable to Azure platform documentation or a formal FR/NFR
`[Decided]`	Explicit design choice in the Decision Log (D1–D11)
`[Derived]`	Logical consequence of a documented constraint or decision
`[Assumed]`	Architectural judgement pending POC validation

Related ADRs

- **ADR-001 — Region Selection** (`acrme_adr_001_region_selection.md`)
- **ADR-002 — Quota and Capacity Management** (`acrme_adr_002_quota_and_capacity_management.md`)
- **ADR-003 — Capacity Management during Disaster Recovery (DR)** (`acrme_adr_003_capacity_manag`

Document Status: Accepted
Next Review: After POC-30 (Quota Groups GA) and POC-31 (quota release latency), and on resolution of G-14 / G-15.